Population Structure: towards bio-geography

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École Nationale Supérieure Agronomique de Toulouse – EcoLab

International Course
Plant Quantitative Genetics, GWAS & Genomic Prediction

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Population structure using SNP data

Outline :

- "Admixture" and using Admixture results

- Multivariate methods :
  - PCA
  - K-means
  - Discriminant Analysis of PCA (DA-PCA)
  - Visualizing pops. on a geographic map (basic)

Why looking for population structure ?

- Knowledge of the genetic structure of the species as basic science result
- Core-collections
- Remove confounding in GWAS and landscape genomics analyses

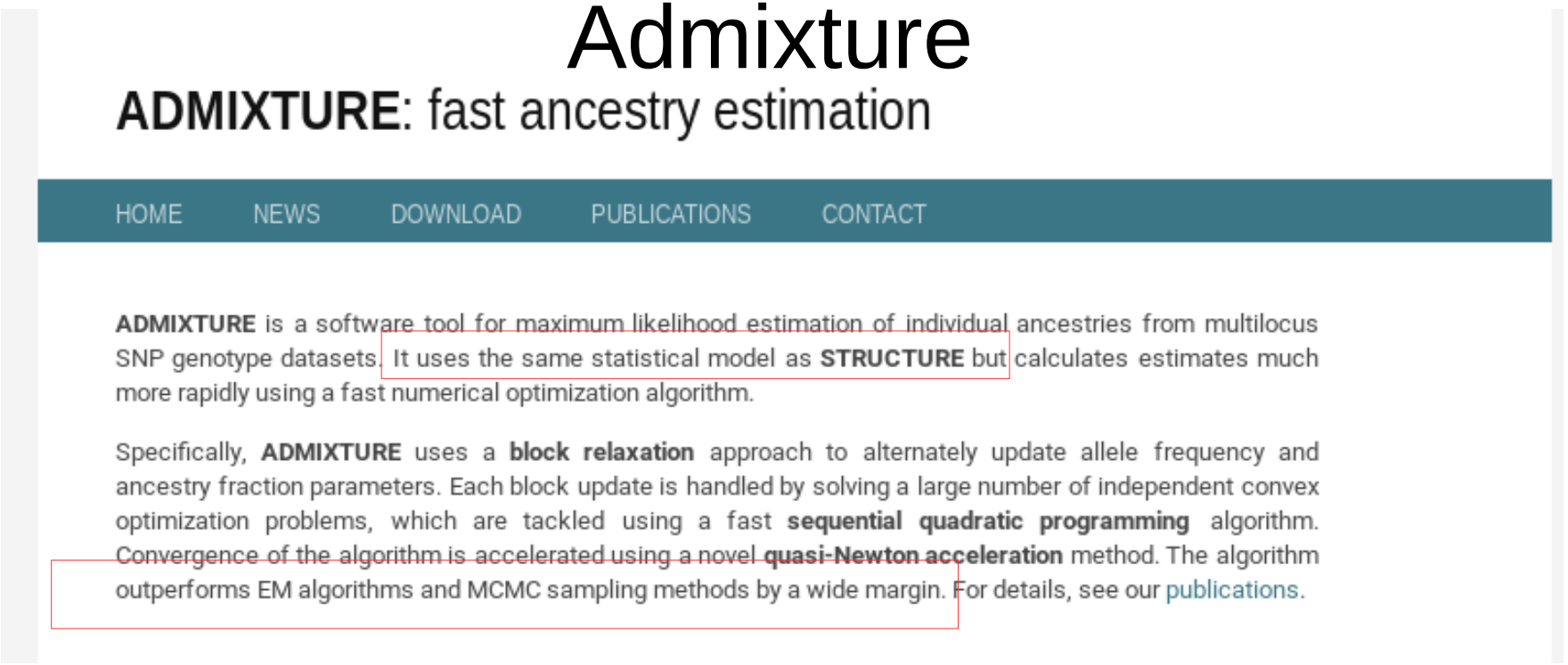
Tools for looking for population structure ?

- Genetic markers : AFLPs, SSRs, SNPs (re-sequencing or GBS or RAD-seq or ….)
- Softwares and computing facilities

Our dataset(s) : Medicago truncatula SNPs

- Two available data-sets for the course :
  - 240K SNPs i.e. 30K SNPs per chromosome
    - Not all computations can be done during the practicals (but can be done at night by your own). raw results will be provided for the course.
  - 16K SNPs i.e. 2K SNPs per chromosome
    - For those who terribly wants to do the computations during the course
- Format of the data sets :
  - Admixture requires "`ADMIXTURE's input is binary PLINK (.bed), ordinary PLINK (.ped), or EIGENSTRAT (.geno) formatted files and its output is simple space-delimited files containing the parameter estimates`."
  - AdeGenet requires "`Data with PLINK format (.raw) can be imported into genlight objects using read.PLINK`".
  - More information on exporting from PLINK can be found at http://zzz.bwh.harvard.edu/plink/dataman.shtml
  - Learn how to play between the different formats for SNPs data !

Inferring population structure using Admixture



- How do I choose the correct value for K? :
  Use ADMIXTURE's cross-validation procedure. A good value of K will exhibit a low cross-validation error compared to other K values.
- https://www.genetics.ucla.edu/software/admixture/
- Admixture is NOT available for Windows

“Admixture”

Resource

Fast model-based estimation of ancestry in unrelated individuals

David H. Alexander,[1,4] John Novembre,[2] and Kenneth Lange[3]

[1]Department of Biomathematics, University of California at Los Angeles, Los Angeles, California 90095, USA; [2]Department of Ecology and Evolutionary Biology, University of California at Los Angeles, Los Angeles, California 90095, USA; [3]Department of Human Genetics and Department of Statistics, University of California at Los Angeles, Los Angeles, California 90095, USA

Our approach is similar to structure’s. Both programs model the probability of the observed genotypes using ancestry proportions and population allele frequencies. Like structure, ADMIXTURE simultaneously estimates population allele frequencies along with ancestry proportions.

mates are directly comparable to those from *structure* and EIGENSTRAT. Taken together, our results show that ADMIXTURE’s computational speed opens up the possibility of using a much larger set of markers in model-based ancestry estimation and that its estimates are suitable for use in correcting for population stratification in association studies.

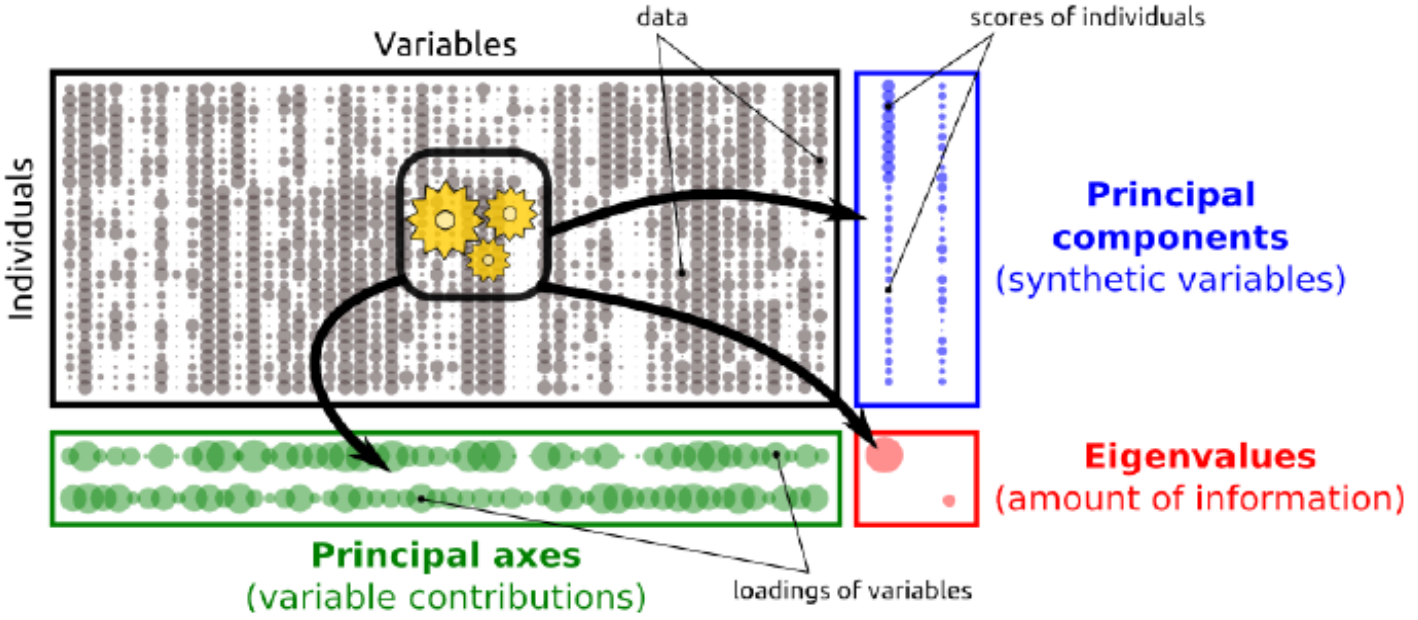
Using results of Admixture analysis

- Analysis was done using the 240K SNPs dataset, for successive values of K = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15
- There is an output file for each parameter set: Q (the ancestry fractions), and P (the allele frequencies of the inferred ancestral populations)
- The Cross-Validation percentage was computed using five-fold procedure, and will be used as the criteria to chose among the putative K values.

- ‘At your keyboards !’

Using multivariate methods to describe diversity

Outputs of multivariate analyses: an overview



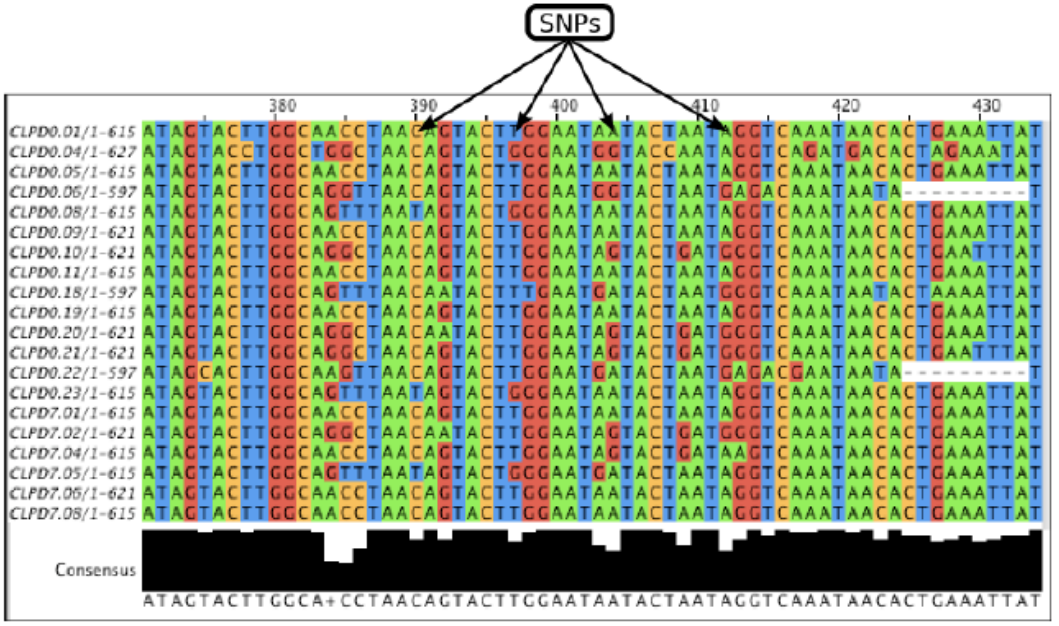
Main outputs:

- **principal components**: diversity amongst individuals
- **principal axes**: nature of the structures
- **eigenvalues**: magnitude of structures

Thibaut Jombart*, Emmanuel Paradis 2015

Using multivariate methods to describe diversity

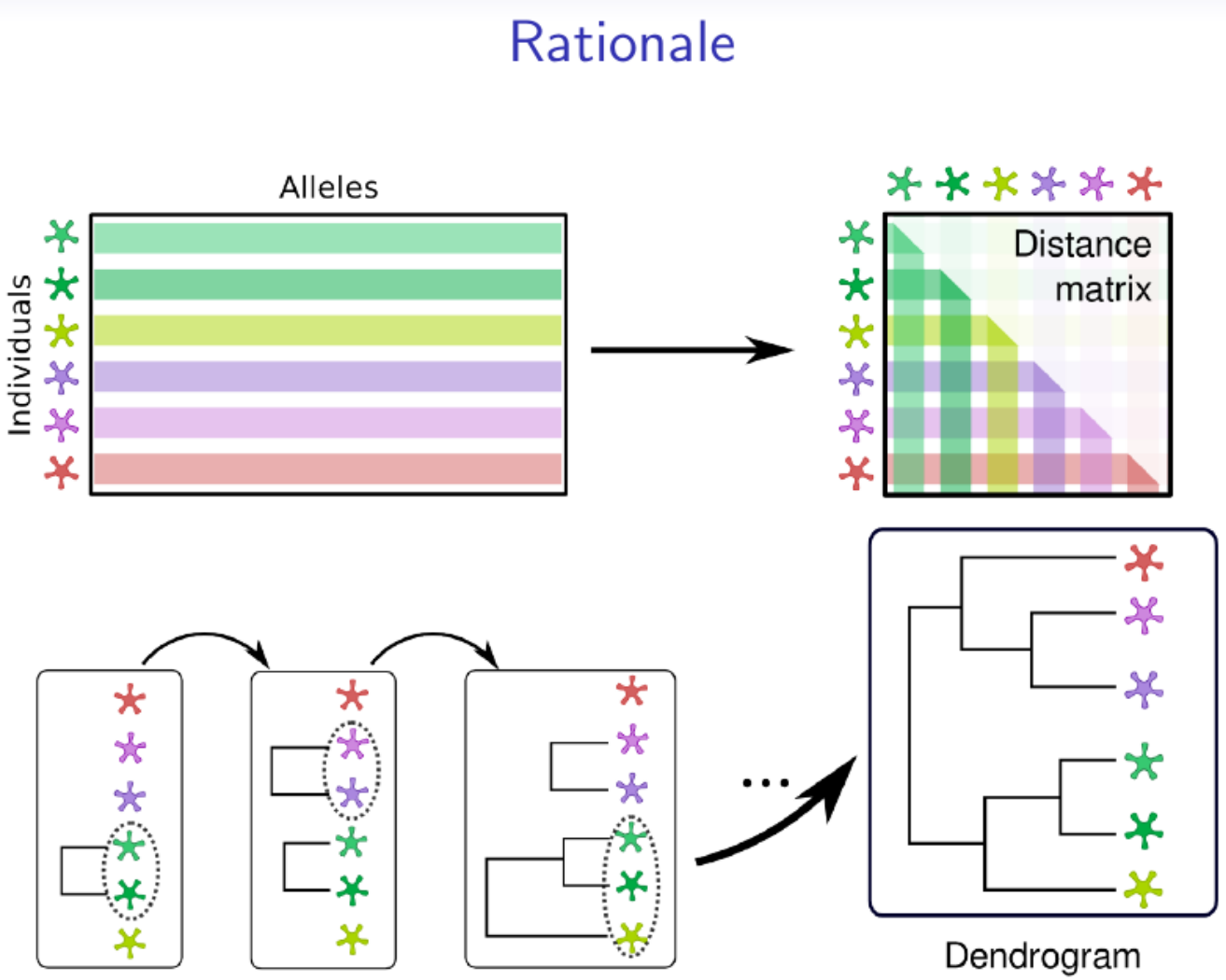
DNA sequences: a rich source of information



- hundreds/thousands individuals
- up to millions of single nucleotide polymorphism (SNPs)

⇒ **Multivariate analysis use to summarize genetic diversity.**

Clustering : a simple idea

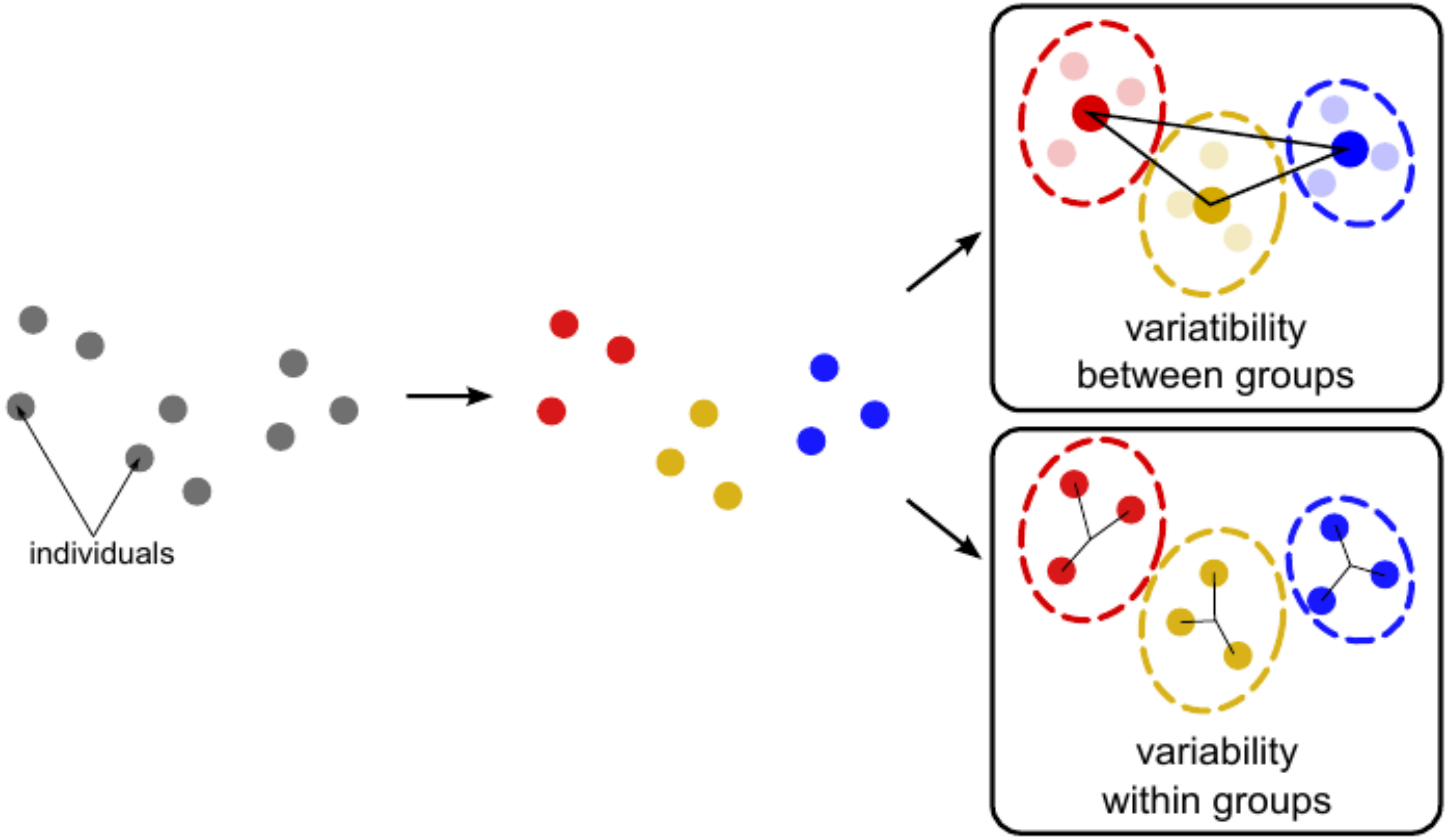


- ‘At your keyboards !’

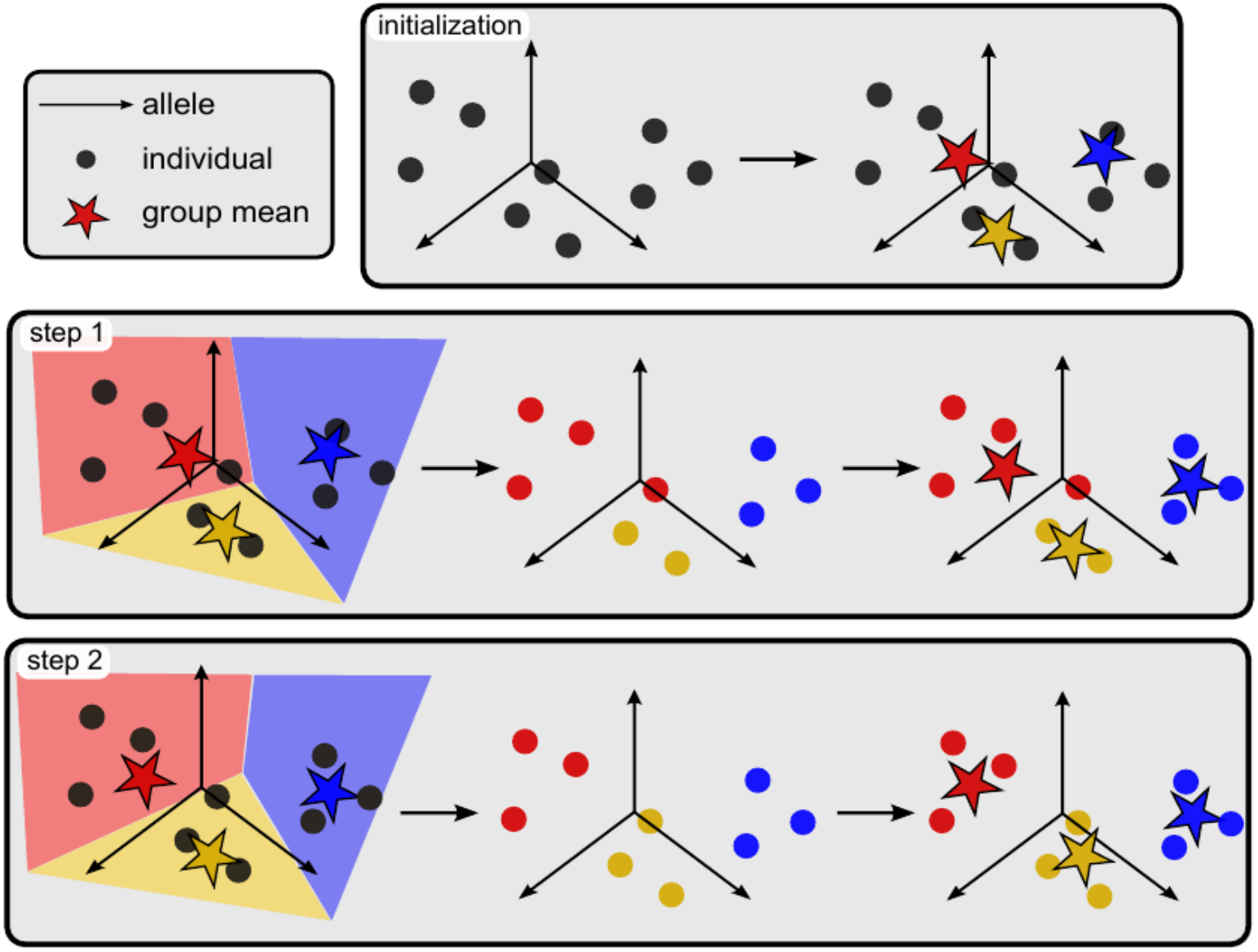
K-means underlying model

ANOVA model:

$$\text{total var.} = (\text{var. between groups}) + (\text{var. within groups})$$

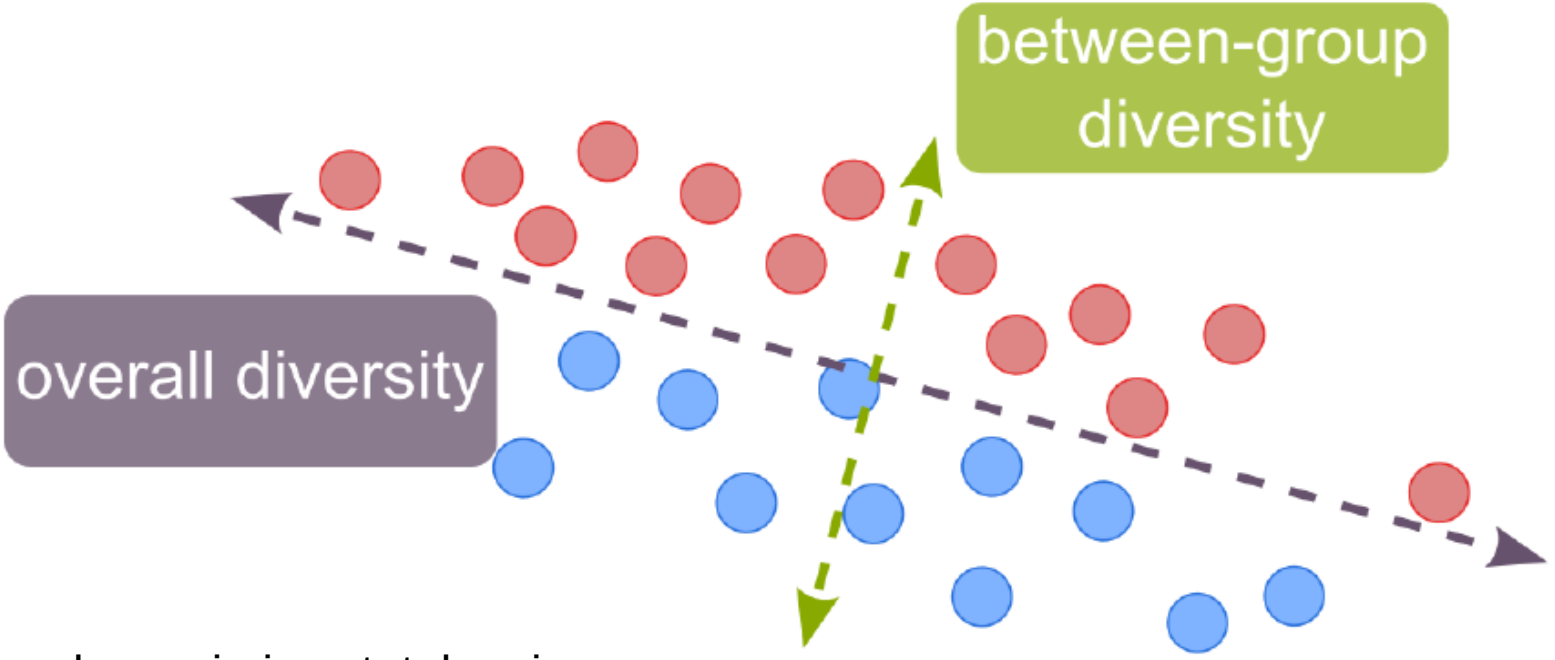


K-means algorithm



- 'At your keyboards !'

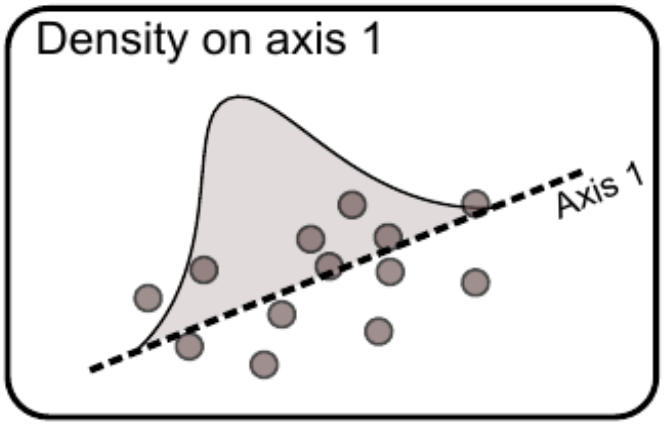
Total variance may not reflect group differences

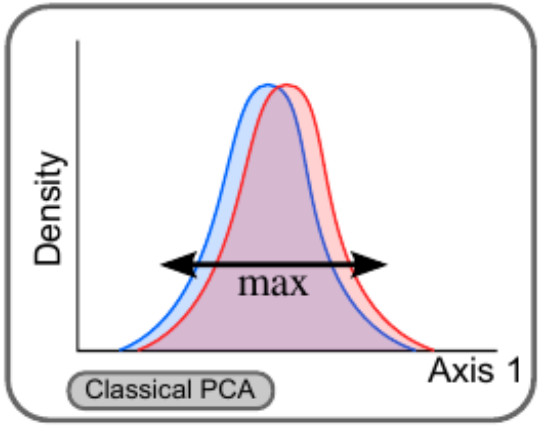


PCA only maximises total variance

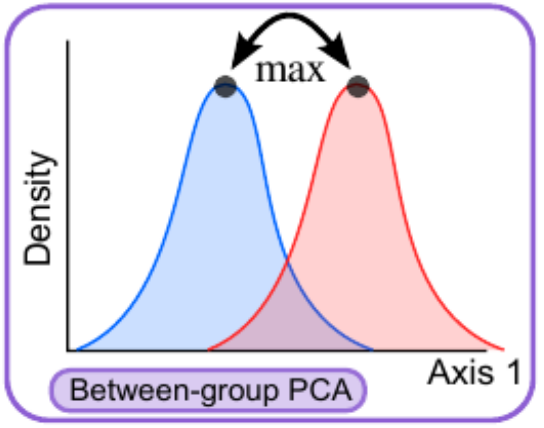
Need to optimize different criteria.

Increasing group differentiation

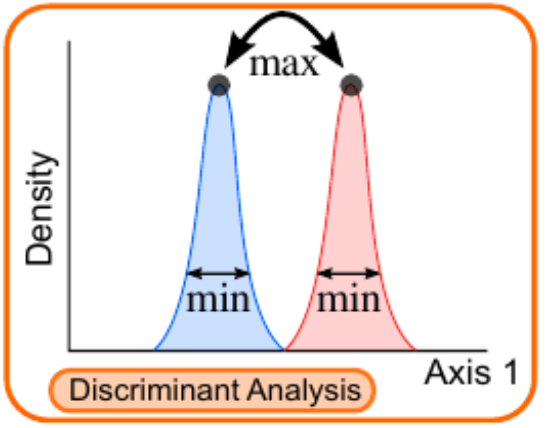




Max. total diversity



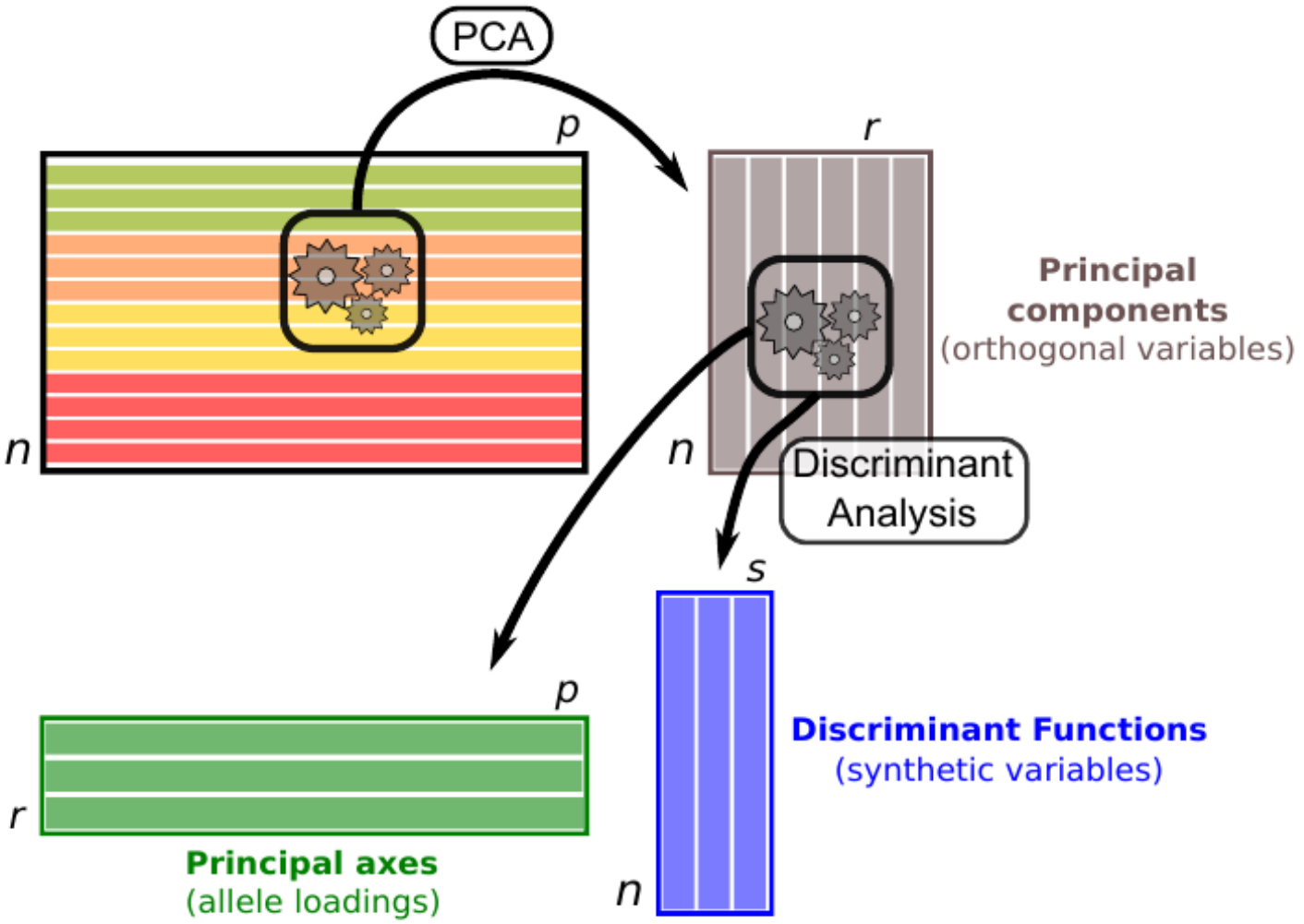
Max. diversity between groups



Max. separation of groups

BUT the number of groups shall be known

Rationale of DAPC



- ‘At your keyboards !’